With **neat diagrams**, describe the image reconstruction using CT.

Image reconstruction in CT is a mathematical process that generates tomographic images from X-ray projection data acquired at many different angles around the patient. It has fundamental impacts on image quality and radiation dose.

CT Principle:

CT uses digital processing to construct 3D internal views from a series of 2D X-rays rotated around an axis. **Backprojection**, the simplest reconstruction method, generates images by mapping measured ray data back to individual voxels.

Projection and Radon Transform

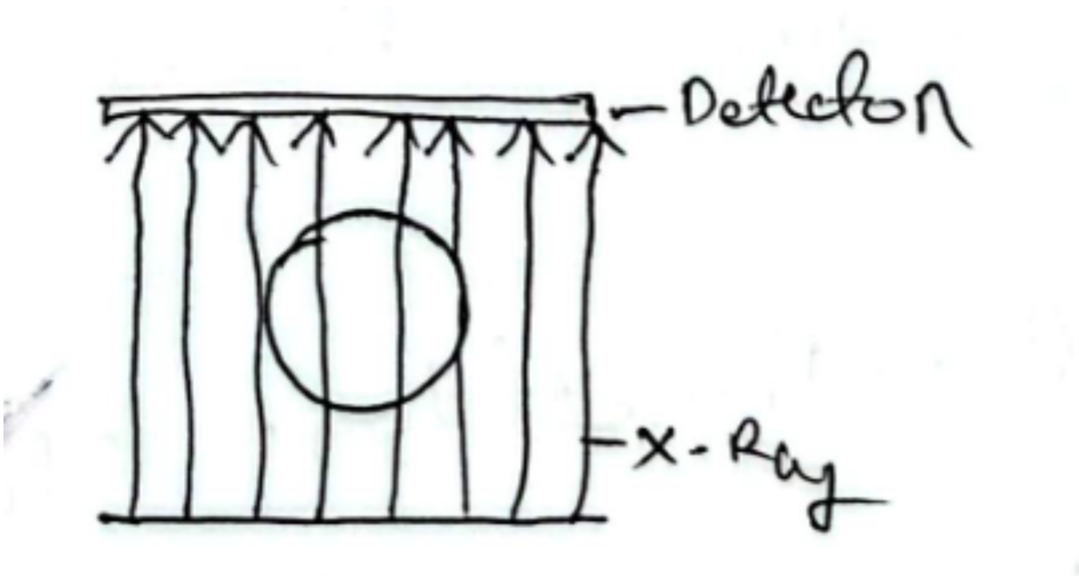
X-rays pass through the object, and the detector measures the loss of intensity after attenuation. A projection is a set of line integrals obtained from the object at a particular angle.

The Radon Transform is the mathematical foundation used to represent these projections. It converts a 2D image into projection data collected from multiple angles.

All projections are then stored in a 2D image called a sinogram.

X-axis: projection angle θ
Y-axis: detector position ρ

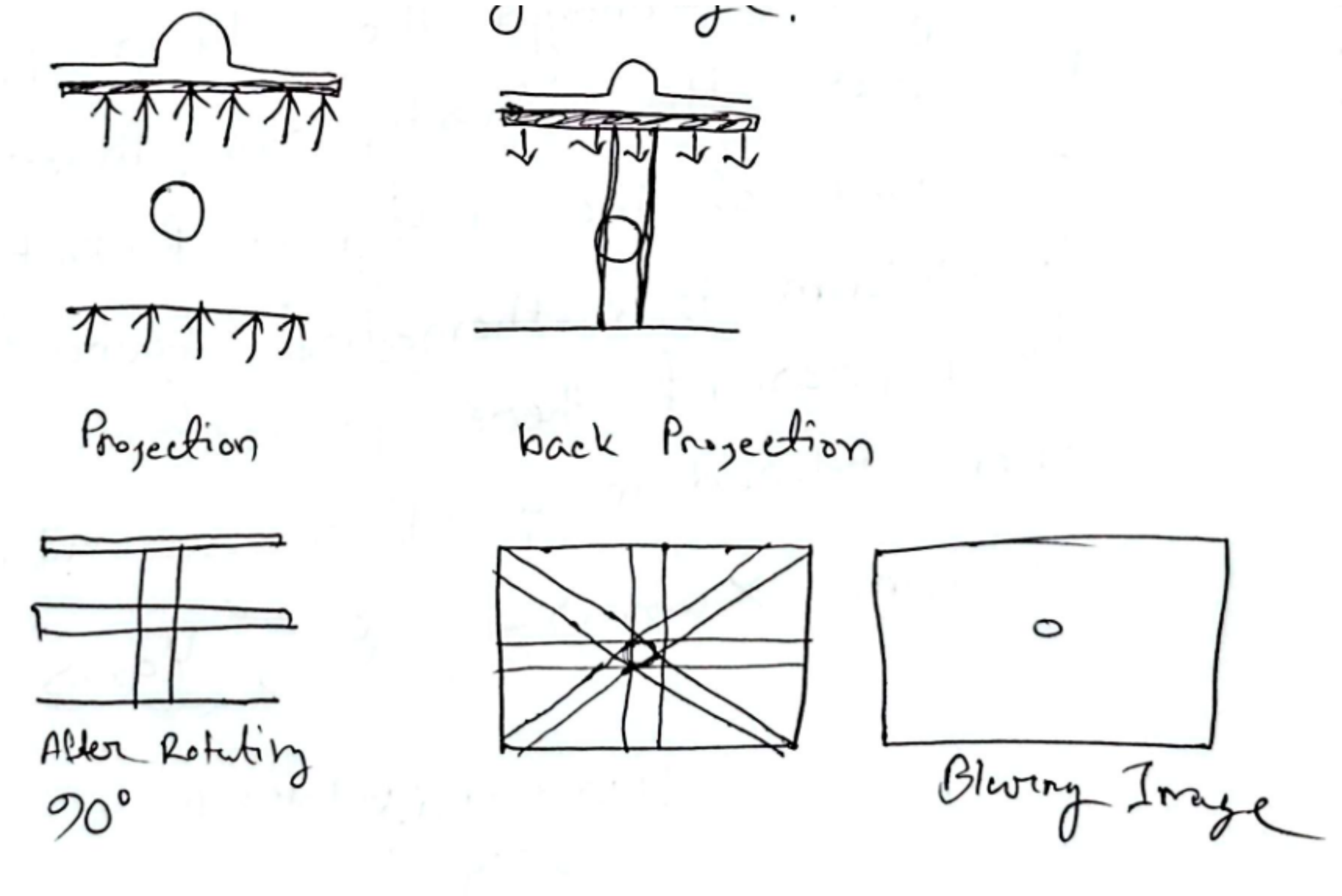
The sinogram contains all the information required for image reconstruction.



Back Projection

The simplest image reconstruction method is backprojection.
In this method, each projection is spread back along the same direction from which it was acquired. When multiple projections are backprojected, their intersections gradually form the reconstructed image.

However, simple backprojection produces a blurred image because projection values are smeared across the image.



Filtered Back Projection (FBP)

is used to reduce the blurring caused by simple backprojection. First, projections are collected from different angles and filtered to sharpen the data. Then, backprojection is applied to the filtered projections, and all projections are summed to form the final reconstructed CT image.FBP produces a clearer and more accurate image than simple backprojection.